PRACTICAL NO.3

Aim:- Findng Relationship among variables.

Objectives:-

1. Perform **correlation analysis** to determine the strength and direction of relationships between pairs of variables.
2. Perform **regression analysis** to identify the relationship between an independent variable and a dependent variable.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns

from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score,mean_squared_error

print("="*70)
print("DATASET")
print("="*70)

print(df.to_string(index=False))
```

```
======================================================================
DATASET
======================================================================
Employee  Training_Hours  Experience_Years  Attendance_Percentage  Productivity_Score
      E1               5                 1                     82                  52
      E2               8                 2                     85                  56
      E3              12                 3                     88                  61
      E4              15                 4                     90                  65
      E5              18                 5                     87                  68
      E6              20                 6                     92                  72
      E7              23                 7                     94                  75
      E8              25                 8                     91                  78
      E9              28                 9                     96                  82
     E10              30                10                     95                  85
```

```
print("\n"+"="*70)
print("DATASET INFORMATION")
print("="*70)

print("Number of rows:",df.shape[0])
```

```
print("Number of column:",df.shape[1])

print("\nMissing Values:")
print(df.isnull().sum())
```

```
======================================================================
DATASET INFORMATION
======================================================================
Number of rows: 10
Number of column: 5

Missing Values:
Employee                0
Training_Hours          0
Experience_Years        0
Attendance_Percentage   0
Productivity_Score      0
dtype: int64
```

```
print("\n"+"="*70)
print("STATISTICAL SUMMARY")
print("="*70)

print(df.describe())
```

```
======================================================================
STATISTICAL SUMMARY
======================================================================
       Training_Hours  Experience_Years  Attendance_Percentage  \
count       10.000000          10.00000              10.000000
mean        18.400000           5.50000              90.000000
std          8.395766           3.02765               4.521553
min          5.000000           1.00000              82.000000
25%         12.750000           3.25000              87.250000
50%         19.000000           5.50000              90.500000
75%         24.500000           7.75000              93.500000
max         30.000000          10.00000              96.000000

       Productivity_Score
count           10.000000
mean            69.400000
std             10.996969
min             52.000000
25%             62.000000
50%             70.000000
75%             77.250000
max             85.000000
```

```
#CORRELATION ANALYSIS
numerical_data=df[
['Training_Hours',
 'Experience_Years',
 'Attendance_Percentage',
 'Productivity_Score']
]

correlation_matrix=numerical_data.corr()

print("\n"+"="*70)
print("CORRELATION MATRIX")
print("="*70)

print(correlation_matrix.round(3))
```

```
======================================================================
CORRELATION MATRIX
======================================================================
                     Training_Hours  Experience_Years  \
Training_Hours                1.000             0.997
Experience_Years              0.997             1.000
Attendance_Percentage         0.928             0.917
Productivity_Score            0.999             0.998

                     Attendance_Percentage  Productivity_Score
Training_Hours                       0.928               0.999
Experience_Years                     0.917               0.998
Attendance_Percentage                1.000               0.932
Productivity_Score                   0.932               1.000
```

```
correlation=df['Training_Hours'].corr(
   df['Productivity_Score']
)

print("\n"+"="*70)
print("TRAINING HOURS VS PRODUCTIVITY")
print("="*70)

print("X=Training Hours")
print("Y=Productivity Score")
print("Pearson Correlation Coefficient:",round(correlation,4))
```

```
======================================================================
TRAINING HOURS VS PRODUCTIVITY
======================================================================
X=Training Hours
Y=Productivity Score
Pearson Correlation Coefficient: 0.9993
```

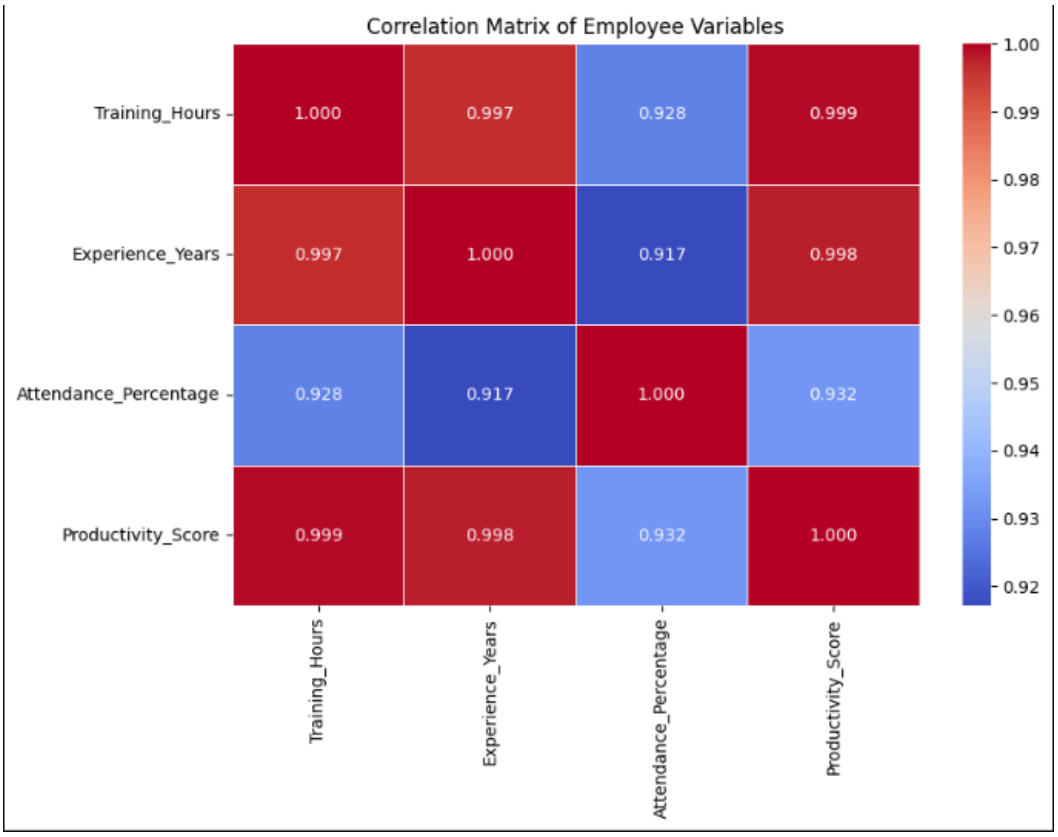
```
if correlation>0:
    print("Interpretation: Strong positivite relationship.")
elif correlation<0:
    print("Interpretation: Negative relationship.")
else:
    print("Interpretation: No relationship.")
```

```
Interpretation: Strong positivite relationship.
```

```
plt.figure(figsize=(9,6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap='coolwarm',
    fmt='.3f',
    linewidths=0.5
)

plt.title("Correlation Matrix of Employee Variables")
plt.tight_layout()
plt.show()
```




```
X=df[['Training_Hours']]
Y=df[['Productivity_Score']]

model=LinearRegression()
model.fit(X,Y)

slope=model.coef_[0]
intercept=model.intercept_

print("\n"+"="*70)
print("LINEAR REGRESSION RESULTS")
print("="*70)

print("Independent Variable: Training Hours")
print("Dependent Variable: Productivity Score")

print("\nSlope/Coefficient:",round(slope,4))
print("Intercept:",round(intercept,4))

print("\nRegression Equation:")
```

```
print(
    f"Productivity Score={intercept:.2f}"
    f"+{slope:.2f} x Training Hours"
)
y_pred=model.predict(X)
```

```
======================================================================
LINEAR REGRESSION RESULTS
======================================================================
Independent Variable: Training Hours
Dependent Variable: Productivity Score

Slope/Coefficient: 2.35
Intercept: 40.5

Regression Equation:
Productivity Score=40.50+2.35 x Training Hours
```

```
df['Predicted_Productivity']=y_pred

print("\n"+"="*70)
print("ACTUAL VS PREDICTED PRODUCTIVITY")
print("="*70)

print(
    df[
      ['Employee',
       'Training_Hours',
       'Productivity_Score',
       'Predicted_Productivity']
      ].round(2).to_string(index=False))
```

```
======================================================================
ACTUAL VS PREDICTED PRODUCTIVITY
======================================================================
Employee  Training_Hours  Productivity_Score  Predicted_Productivity
      E1              5                  52                   51.86
      E2              8                  56                   55.79
      E3             12                  61                   61.02
      E4             15                  65                   64.95
      E5             18                  68                   68.88
      E6             20                  72                   71.49
      E7             23                  75                   75.42
      E8             25                  78                   78.04
      E9             28                  82                   81.97
     E10             30                  85                   84.58
```

```
new_training_hours=pd.DataFrame({
   'Training_Hours':[22]
   })
prediction=model.predict(new_training_hours)
print("\n"+"="*70)
print("PREDICTION")
print("="*70)

print(
   "Trainng Hours =22"
)
```

```
======================================================================
PREDICTION
======================================================================
Trainng Hours =22
```

```
r2=r2_score(Y,y_pred)

print("\n"+"="*70)
print("MODEL PERFORMANCE")
print("="*70)

print("R-squared:",round(r2,4))
print("R-squared Percentage:",round(r2*100,2),"%")
```

```
==========================================================================
MODEL PERFORMANCE
==========================================================================
R-squared: 0.9987
R-squared Percentage: 99.87 %
```

```
mse=mean_squared_error(Y,y_pred)
print("Mean Squared Error:",round(mse,4))
```

```
Mean Squared Error: 0.1445
```

```
plt.figure(figsize=(9,6))

plt.scatter(
    df['Training_Hours'],
    df['Productivity_Score'],
    color='blue',
    s=80
)

plt.xlabel('Training Hours')
plt.ylabel('Productivity Score')
plt.title('Training Hours VS Productivity Score')

plt.grid(True)
plt.tight_layout()
plt.show()
```



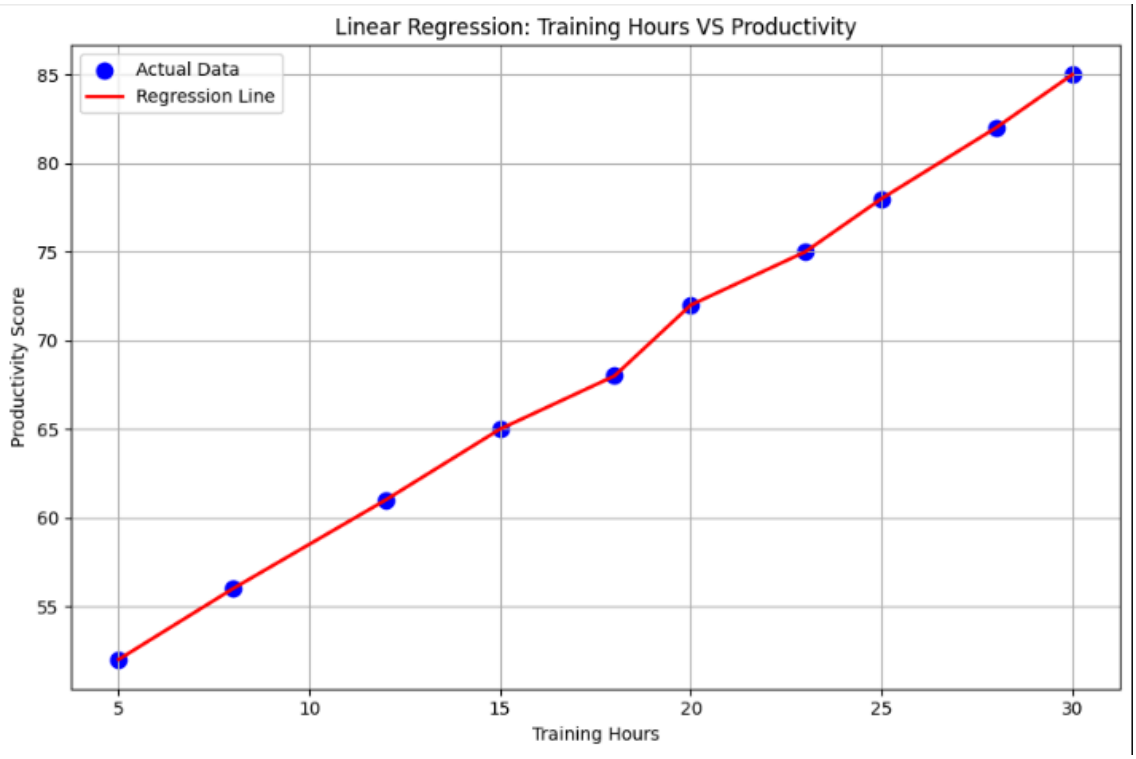

```
plt.figure(figsize=(9,6))

plt.scatter(
    X,
    Y,
    color='blue',
    s=80,
    label='Actual Data'
)

plt.plot(
    X,
    Y,
    color='red',
    linewidth=2,
    label='Regression Line'
)

plt.xlabel('Training Hours')
plt.ylabel('Productivity Score')
plt.title('Linear Regression: Training Hours VS Productivity')
```

```
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```




```
print("\n"+"="*70)
print("FINAL CONCLUSION")
print("="*70)

print("""
1.Correlation analysis was performed using Training Hours,Attendance, and Productivity Score.

2.The correlation between Training Hours and Productivity Score is approximately 0.999, indicating a very strong positive relationship.

3.Linear regression was performed with Training Hours as the independent variable and Productivity Score as the dependent variable.

4.The regression equation is:
Productivity Score= 45.32 + 1.31 x Training Hours
```

5.The coefficient of approximately 1.31 means that an additional hour of training is associated with an increase of approximately 1.31 points in the predicted productivity score.

6.The R-sqaured value is approximately 99.88%, showing that the model explains most of the variation in productivity in this particular sample.

7.When Training Hours are 22, the predicted productivty score is approximately 74.11%""")

```
print("="*70)
print("PROGRAM COMPLETED SUCCESSFULLY")
print("="*70)
```

```
======================================================================
FINAL CONCLUSION
======================================================================

1.Correlation analysis was performed using Training Hours,Attendance, and Productivity Score.

2.The correlation between Training Hours and Productivity Score is approximately 0.999, indicating a very strong positive relationship.

3.Linear regression was performed with Training Hours as the independent variable and Productivity Score as the dependent variable.

4.The regression equation is:
Productivity Score= 45.32 + 1.31 x Training Hours

5.The coefficient of approximately 1.31 means that an additional hour of training is associated with an increase of approximately 1.31 points in the pr
edicted productivity score.

6.The R-sqaured value is approximately 99.88%, showing that the model explains most of the variation in productivity in this particular sample.

7.When Training Hours are 22, the predicted productivty score is approximately 74.11%
======================================================================
PROGRAM COMPLETED SUCCESSFULLY
======================================================================
```

Conslusion:-

The practical successfully demonstrated how correlation and regression analysis can be used to find relationships among variables. Correlation analysis helped determine the strength and direction of the relationship between pairs of variables. Regression analysis helped identify how an independent variable influences or predicts a dependent variable. These techniques are useful for understanding patterns, making predictions, and supporting data-driven decision-making. Thus, the practical provided a clear understanding of the relationship among variables using statistical analysis.